Fair Atomic Governance: Allocating Decision Boundaries under Shared Resource Constraints in Multi-Agent Systems

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Abstract

Atomic decision boundaries guarantee that each agent action is evaluated and committed in a single indivisible step, eliminating the time-of-check/time-of-use gap that undermines split evaluation systems. This correctness guarantee, however, operates *locally*: it determines whether a given state transition is admissible, but not *which agent* among many competing requestors reaches the boundary to attempt that transition. When multiple agents share governance infrastructure, per-agent bounded enforcement induces a global capacity $K = N \cdot k_0$ that scales linearly with agent count. This creates a structural vulnerability—*Sybil amplification*—in which an actor controlling m agents captures $m \cdot k_0$ admissible actions without violating any local constraint.

We extend the labeled transition system (LTS) model of atomic governance to the multi-agent setting, separating the correctness of individual admission decisions from the fairness of how those decisions are distributed across agents and actors. We formalize three classes of allocation-level failure—Sybil amplification, temporal domination, and resource contention unfairness—and introduce a hierarchy of fairness notions suited to atomic governance: share fairness, actor-level proportionality, envy-freeness, and strategy-proofness.

We prove (Theorem 5.1) that, under per-agent bounded enforcement without actor-level constraints, actor-level fairness cannot be guaranteed regardless of how admission decisions are made. We prove (Theorem 5.3) that admission control alone—even when atomically correct—is insufficient to guarantee fair allocation: fairness requires an explicit allocation layer above the decision boundary. We prove (Theorem 5.4) that strategy-proofness and actor-level proportionality are jointly unachievable under pure per-agent enforcement, formalizing the cost of identity fragmentation.

We analyze four allocation mechanisms—per-agent token bucket, round-robin fair queuing, actor-aware rate limiting, and weighted fair queuing— establishing which fairness properties each mechanism satisfies and where each fails. We further characterize the interaction of allocation fairness with the Escalate outcome and with the Invariant Measurement Layer, showing that allocation bias induces a form of behavioral drift detectable above the enforcement boundary.

These results close the governance quartet: correctness at the decision boundary (Fernandez [5], Fernandez [6]), behavioral invariance above it (Fernandez [7]), and fairness in how the boundary is accessed across competing agents (this paper).

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1 Introduction

Three prior works in this series have established a layered governance architecture for autonomous agent systems. Fernandez [5] (Paper 0) proves that admissibility can only be guaranteed at execution time if the decision and the resulting state transition are jointly determined as a single indivisible step in the LTS—the *atomic decision boundary*. Fernandez [6] (Paper 1, ACP) instantiates this boundary as a concrete protocol: each agent action acquires a single-use Execution Token emitted and consumed atomically with the ledger state mutation, ensuring that the state against which admissibility is evaluated is precisely the state against which the transition commits. Fernandez [7] (Paper 2, IML) operates *above* the atomic boundary and detects behavioral drift that accumulates within the compliance region $g^{-1}(0)$—sequences of individually admissible actions that collectively violate admission-time behavioral invariants.

Each of these layers addresses a question of *correctness*: whether an individual action is admissible at the moment it fires (Papers 0–1), and whether an agent's behavioral trajectory remains within its authorized scope (Paper 2). What none of these layers addresses is a question of *distribution*: when multiple agents simultaneously compete for the governance infrastructure that evaluates their requests, which agents get to present their requests to the atomic boundary, and when? This is the question of *allocation fairness*, and it is the subject of the present paper.

The contention problem. Consider a shared execution environment governed by an atomic admission control mechanism with per-agent bound k_0. In isolation, each agent a_i can execute at most k_0 admissible actions before reaching its enforcement threshold. With N agents, the global admissible capacity is $K = N \cdot k_0$. Now suppose a single actor controls m of those N agents. Because enforcement is *per-agent*, each of the actor's m agents accumulates its counter independently. The actor can therefore execute up to $m \cdot k_0$ actions—a share of m/N of the total capacity—without any of its individual agents exceeding k_0. The system is atomically correct at every decision point: every action was individually admissible when it fired. Yet the allocation is concentrated, and potentially captures the entire global capacity if $m = N$.

We call this phenomenon *Sybil amplification*: the ability of an actor to linearly scale its effective governance capacity by fragmenting its identity across multiple agents, each individually compliant. The name derives from the Sybil attack in peer-to-peer systems [4], where a single adversary creates multiple identities to subvert reputation or voting mechanisms. Here the same structural vulnerability applies to admission-controlled governance: per-identity enforcement, without cross-identity aggregation, is exploitable by identity multiplication.

Example 1.1 (Sybil amplification under ACP)**.** Under ACP [6], Experiment 13 establishes the *Coordination Window*: N agents issuing homogeneous requests on a shared resource each accumulate risk independently via agent-scoped $PatternKey(agentID, capability, resource)$, yielding a total coordination window $\mathrm{CW}_{\mathrm{appr}} = 2N$ with $k_0 = 2$. This linear scaling is a design property, not a flaw: ACP correctly ensures each agent is governed within its local context. The gap identified in the present paper is that a single actor controlling all N agents can exploit this window to capture $2N$ admissible actions before any escalation—precisely the entire system capacity—while every local enforcement decision is correct.

Sybil amplification is the most structurally sharp of three failure modes we identify at the allocation layer. The second is *temporal domination*: in a resource-constrained system, agents that arrive or submit requests earlier consume available capacity, leaving later agents without execution opportunities regardless of their individual bounds. The third is *resource contention unfairness*: when multiple agents compete for the same resource, the absence of an allocation policy allows

random or structural biases—request timing, retry aggressiveness, network proximity—to produce skewed distributions of admissible actions without violating any governance rule.

Correctness is local; fairness is global. The central observation driving this paper is a structural separation: *correctness is a property of individual transitions; fairness is a property of the distribution of transitions across agents.* The atomic decision boundary (Paper 0) ensures that, at the moment a transition fires, the state justifies the ALLOW decision. This is a *local* guarantee—it concerns a single $(s, a) \mapsto (d, s')$ evaluation. Fairness, by contrast, concerns the *aggregate pattern* of which agents' requests reach the boundary, at what rate, and with what outcomes. It is a *global* property, invisible to any individual decision point.

This separation implies that a system can be perfectly correct—every decision boundary is atomic, every ALLOW is justified—while being arbitrarily unfair in how it distributes those allowances across agents. Conversely, a fair system that is not atomically correct cannot guarantee that the actions it allocates are actually admissible. The two properties are orthogonal dimensions of governance, and both are necessary.

The allocation layer. We define the *allocation layer* as the mechanism that determines which agent's pending admissible request is presented to the atomic decision boundary at each step. The allocation layer sits between the agent population and the governance boundary: it is the scheduler that decides *who goes next.* Its decisions are invisible to the admission control function $F : S \times A \to \mathcal{D} \times S$ of Paper 0, which only sees the (s, a) pair presented to it. Yet those scheduling decisions determine the system-wide distribution of admissible actions and thus all fairness properties of the deployment.

The allocation layer is not hypothetical. In any multi-agent system with a shared admission control endpoint—whether a database with serializable transactions, an API gateway with rate limiting, or an ACP ledger with atomic token issuance—multiple concurrent requests must be serialized at the boundary. The order in which they are serialized is always determined by some mechanism: a FIFO queue, a priority scheme, a random tie-break, or first-come-first-served. We make this implicit mechanism explicit, define its fairness properties formally, and analyze the consequences of different choices.

Contributions.

(i) We extend the LTS-based governance model of Fernandez [5] to the multi-agent setting, introducing the agent population $\mathcal{A}$, the actor mapping $\pi : \mathcal{A} \to \mathcal{U}$, per-agent bounds k_0, and the allocation function α as a first-class formal object (§2).

(ii) We formalize a hierarchy of fairness notions for atomic governance: share fairness, actor-level proportionality, envy-freeness, and strategy-proofness, relating them to classical fairness theory (§3).

(iii) We identify and formally characterize three allocation-level failure modes—Sybil amplification, temporal domination, and resource contention unfairness—showing each is consistent with full atomic correctness (§4).

(iv) We prove the **Sybil Amplification Theorem** (Theorem 5.1): under per-agent bounded enforcement without actor-level constraints, actor-level share grows linearly with the number of agents controlled, and actor-level fairness cannot be guaranteed (§5).

(v) We prove the **Allocation Necessity Theorem** (Theorem 5.3): a system satisfying the atomic decision boundary property is not sufficient to guarantee fair allocation; an explicit allocation mechanism above the boundary is a necessary additional component (§5).

(vi) We prove the **Strategy-Proofness Impossibility** (Theorem 5.4): under per-agent independent enforcement, no allocation mechanism can simultaneously guarantee actor-level proportionality and strategy-proofness against identity fragmentation (§5).

(vii) We analyze four allocation mechanisms with respect to the fairness hierarchy, establishing which properties each mechanism achieves and under what conditions (§6, §7).

(viii) We characterize the interaction of allocation fairness with the ESCALATE outcome and with the IML, showing that allocation bias induces a drift signal detectable by the invariant measurement layer (§8).

Position in the quartet. This paper is Paper 3 in a series of four. Paper 0 [5] establishes the structural necessity of atomic decision boundaries. Paper 1 [6] (ACP) instantiates those boundaries as a concrete protocol. Paper 2 [7] (IML) monitors behavioral drift above the boundary. The present paper closes the quartet by addressing the allocation layer: given a correctly functioning atomic boundary and a behavioral invariant monitor, how should the governance infrastructure be shared among competing agents so that no actor can disproportionately capture the system's execution capacity? The four layers together define a complete governance architecture: what is admissible (Papers 0–1), how far behavior has drifted from admission-time contracts (Paper 2), and how admissible actions are distributed (Paper 3).

2 System Model

2.1 Multi-Agent Labeled Transition System

We build on the execution model of Fernandez [5]. Let $M = (S, \mathrm{Act}, \rightarrow)$ be an LTS as in Definition 2.1 of that work, where $\mathrm{Act} = A_{\mathrm{ag}} \cup A_{\mathrm{env}}$ is partitioned into agent actions and environment actions. We now make the agent population explicit.

Definition 2.1 (Agent Population)**.** An *agent population* is a finite set $\mathcal{A} = \{a_1, \ldots, a_N\}$. Each agent $a_i \in \mathcal{A}$ generates action requests from a per-agent action alphabet $\mathrm{Act}_i \subseteq A_{\mathrm{ag}}$, with $A_{\mathrm{ag}} = \bigcup_{i=1}^{N} \mathrm{Act}_i$. We write $a_i : x$ for the action request of agent a_i for action $x \in \mathrm{Act}_i$.

Definition 2.2 (Actor Mapping)**.** An *actor set* is a finite set $\mathcal{U} = \{u_1, \ldots, u_M\}$ with $M \leq N$. The *actor mapping* $\pi : \mathcal{A} \rightarrow \mathcal{U}$ assigns each agent to its controlling actor. For actor $u_j \in \mathcal{U}$, we write $\mathcal{A}_j = \pi^{-1}(u_j) = \{a_i \mid \pi(a_i) = u_j\}$ for the set of agents it controls, with $|\mathcal{A}_j| = m_j$. The partition $\{\mathcal{A}_j\}_{j=1}^{M}$ covers $\mathcal{A}$ exactly.

Remark 2.3 (Observability of π)*.* The actor mapping π may or may not be observable to the governance system. In the base model (§2), we make no observability assumption; we distinguish two regimes in §6: *identity-oblivious* mechanisms that do not use π, and *actor-aware* mechanisms that do. The Sybil amplification vulnerability (§4) arises precisely in identity-oblivious systems.

2.2 Atomic Decision Boundary (Multi-Agent)

The atomic decision function $F : S \times A_{\text{ag}} \to \mathcal{D} \times S$ of Fernandez [5] is unchanged. In the multi-agent setting, multiple agents may present requests to F concurrently. Because F is a single indivisible LTS step (Definition 3.3 of Paper 0), concurrent requests must be *serialized*: at most one agent's request is evaluated by F in any given LTS step.

Definition 2.4 (Serialized Boundary)**.** A *serialized atomic decision boundary* for agent population $\mathcal{A}$ is a pair (F, α) where:

(i) $F : S \times A_{\text{ag}} \to \mathcal{D} \times S$ is an atomic decision function as in Fernandez [5], Definition 3.3;

(ii) $\alpha : \text{PendingRequests}(\mathcal{A}) \times S \to \mathcal{A}$ is an *allocation function* that, given the set of currently pending requests and the current state, selects the next agent whose request is presented to F.

The function α is the formal object that determines allocation fairness. The correctness guarantee of F is independent of α: whatever agent α selects, F evaluates its request atomically against the current state.

The key observation is in the final sentence of Definition 2.4: F and α are *independent*. The atomic boundary guarantees local correctness for whichever request it evaluates; the allocation function determines *whose* request is evaluated. Fairness is a property of α, not of F.

2.3 Per-Agent Bounded Enforcement

Following Fernandez [6], each agent operates under a bound $k_0 \in \mathbb{N}$ on the number of admissible actions it may execute within a given *enforcement context* (a time window, a resource scope, or a behavioral pattern, as determined by the *PatternKey* scoping mechanism of ACP).

Definition 2.5 (Admission Counter and Bound)**.** For agent $a_i \in \mathcal{A}$ and enforcement context c, let $\kappa_i(t, c) \in \mathbb{N}$ denote the number of actions executed by a_i in context c up to time t for which F returned Allow. The *per-agent bound* k_0 requires:

$$\kappa_i(t, c) \;\leq\; k_0 \quad \forall\, i,\, t,\, c.$$

When $\kappa_i(t, c) = k_0$, further requests by a_i in context c receive Escalate or Refuse per the risk scoring function of Fernandez [6]. Enforcement of this bound is per-agent and independent: $\kappa_i(t, c)$ is not shared across agents.

Definition 2.6 (Global Capacity)**.** For a population $\mathcal{A}$ of N agents each with bound k_0 in context c, the *global admissible capacity* is:

$$K \;=\; N \cdot k_0.$$

This is the maximum number of Allow decisions that F can issue across all agents before enforcement thresholds are reached.

2.4 Allocation Function and Execution Shares

We now formalize the distribution of admissible actions across agents.

Definition 2.7 (Allocation Function)**.** Let $\mathcal{P}(\mathcal{A})$ denote the set of nonempty subsets of $\mathcal{A}$. An *allocation function* is a mapping:

$$\alpha : \mathcal{P}(\mathcal{A}) \times S \;\to\; \mathcal{A}$$

such that $\alpha(P, s) \in P$ for all (P, s). Here $P \subseteq \mathcal{A}$ is the set of agents with pending requests at state s.

Definition 2.8 (Execution Share)**.** For a finite execution trace σ of length T (counting ALLOW decisions only), the *execution share* of agent a_i is:

$$S_i \;=\; \frac{\kappa_i}{\sum_{j=1}^{N} \kappa_j} \;=\; \frac{\kappa_i}{K'},$$

where κ_i is the count of ALLOW decisions issued to a_i and $K' \leq K$ is the total count across all agents.

Definition 2.9 (Actor Share)**.** For actor $u_j \in \mathcal{U}$, the *actor share* is:

$$S_{u_j} \;=\; \sum_{a_i \in \mathcal{A}_j} S_i.$$

The actor share measures the aggregate allocation captured by all agents under u_j's control.

2.5 Shared Resources and Contention

Definition 2.10 (Resource Set and Contention)**.** Let $\mathcal{R} = \{r_1, \ldots, r_L\}$ be a set of shared resources. Each action $x \in A_{\text{ag}}$ targets a resource $\rho(x) \in \mathcal{R}$. A *contention event* at resource r occurs when $|\{a_i \in \mathcal{A} : a_i \text{ requests } x \text{ with } \rho(x) = r, \text{ simultaneously}\}| > 1$: multiple agents present admissible requests for the same resource in the same time step, requiring α to serialize them.

3 Fairness Definitions

3.1 Share Fairness

Definition 3.1 (Agent-Level Share Fairness)**.** A system satisfies *ε-share fairness* if for all agents $a_i \in \mathcal{A}$:

$$\left| S_i - \tfrac{1}{N} \right| \;\leq\; \varepsilon,$$

for some $\varepsilon \geq 0$. We say the system is *exactly share-fair* when $\varepsilon = 0$, i.e., every agent executes exactly $K/N = k_0$ admitted actions.

Agent-level share fairness is the natural baseline: absent any reason to prefer one agent over another, the global capacity should be divided equally. It corresponds to *proportional fairness* restricted to the uniform-weight case [10].

3.2 Actor-Level Proportionality

Agent-level share fairness is *insufficient* when actors control multiple agents. An actor with m_j agents can achieve share $S_{u_j} = m_j/N$ even when each of its agents receives exactly $1/N$ of capacity. If one actor controls half the population, it captures half the capacity—a disproportionate outcome from the perspective of the M actors sharing the system.

Definition 3.2 (Actor-Level Proportionality)**.** A system satisfies *ε-actor-level proportionality* if for all actors $u_j \in \mathcal{U}$:

$$\left| S_{u_j} - \tfrac{1}{M} \right| \;\leq\; \varepsilon.$$

Remark 3.3 (Incompatibility with agent-level fairness under skewed populations). If actor u_1 controls m_1 agents and actor u_2 controls m_2 agents with $m_1 \neq m_2$, then ε-share fairness and ε-actor-level proportionality are jointly achievable only if k_0 allocations are fractional or agents receive unequal per-agent bounds. Under the uniform-k_0 model, the two notions are compatible only when $m_j = N/M$ for all j—i.e., when all actors control the same number of agents. The tension between them is formalized in Theorem 5.1.

3.3 Envy-Freeness

Definition 3.4 (Envy-Freeness)**.** An allocation $(S_1, \ldots, S_N)$ is *envy-free* if no agent strictly prefers another agent's allocation:

$$\forall\, i, j \in \mathcal{A}: \quad v_i(S_i) \;\geq\; v_i(S_j),$$

where $v_i : [0,1] \to \mathbb{R}$ is agent a_i's valuation function over execution shares. When valuations are identical and monotone (all agents prefer more execution capacity), envy-freeness reduces to $S_i \geq S_j$ for all i, j—i.e., equal shares.

Envy-freeness is a classical fairness concept originating in welfare economics [8]. In the governance setting, it captures the intuition that no agent should be able to observe another agent's allocation and wish they had received it instead. Under uniform valuations, envy-freeness and ε-share fairness coincide in the limit $\varepsilon \to 0$.

3.4 Strategy-Proofness Against Identity Fragmentation

The weakest fairness notions (share fairness, actor-level proportionality) are *incentive-compatible* only if actors cannot benefit by manipulating their agent structure. Strategy-proofness formalizes this.

Definition 3.5 (Strategy-Proofness Against Fragmentation)**.** An allocation mechanism α is *strategy-proof against identity fragmentation* if, for any actor u_j controlling m_j agents, the actor share S_{u_j} is independent of m_j:

$$\frac{\partial\, S_{u_j}}{\partial\, m_j} \;=\; 0.$$

Equivalently, splitting a single agent a_i into two agents $a_i^{(1)}, a_i^{(2)}$ under the same actor does not increase S_{u_j}.

Strategy-proofness is the strongest notion in our hierarchy. It captures the key governance requirement that the admission control system should not create incentives for actors to proliferate agent identities as a means of capturing execution capacity.

3.5 Temporal Fairness and Starvation Freedom

Definition 3.6 (Starvation Freedom)**.** A system is *starvation-free* if every agent with a pending request that is admissible in the current state eventually receives an Allow decision:

$$\forall\, a_i \in \mathcal{A},\ \forall\, x \in \mathrm{Act}_i: \quad \mathrm{Adm}(s, x) = \mathsf{true} \;\Rightarrow\; \exists\, t' > t : F(s_{t'}, x) = (\textsc{Allow}, s'').$$

Starvation freedom is orthogonal to share fairness: a system can provide share fairness in expectation while still causing some agents to starve in adversarial scheduling scenarios. We treat it as a minimum liveness requirement distinct from the distributional fairness properties above.

4 Allocation-Level Failure Modes

4.1 Sybil Amplification

Definition 4.1 (Sybil Amplification)**.** A system exhibits *Sybil amplification* if the actor share S_{u_j} can be increased by increasing $m_j = |\mathcal{A}_j|$ while holding all other parameters fixed:

$$\exists\, u_j \in \mathcal{U} : \quad S_{u_j}(m_j + 1) \;>\; S_{u_j}(m_j).$$

Example 4.2 (Sybil amplification witness)**.** Let $N = 4$ agents with $k_0 = 2$ each, so $K = 8$. Actors are u_1 (controlling $m_1 = 1$ agent) and u_2 (controlling $m_2 = 3$ agents). Under ε-share fairness at $\varepsilon = 0$, each agent receives $k_0 = 2$ admissible actions. The actor shares are $S_{u_1} = 2/8 = 0.25$ and $S_{u_2} = 6/8 = 0.75$. Actor u_2 captures three-quarters of all admissible capacity. No local constraint is violated: each individual agent executes exactly $k_0 = 2$ actions. The system is atomically correct and exactly share-fair, yet actor-level proportionality fails: $S_{u_2} - 1/2 = 0.25 > \varepsilon$ for any reasonable ε.

Now suppose u_2 registers a fourth agent (i.e., $m_2 \to 4$, $N \to 5$, $K \to 10$). Under the same per-agent rule, $S_{u_2} = 8/10 = 0.80$. The actor share has grown by 0.05 with zero change in behavior of any individual agent.

4.2 Temporal Domination

Definition 4.3 (Temporal Domination)**.** A system exhibits *temporal domination* if there exist agents $a_i, a_j \in \mathcal{A}$ and time points $t_1 < t_2$ such that:

(i) a_i submits its first request at time t_1;

(ii) a_j submits its first request at time $t_2 > t_1$;

(iii) $\kappa_i(T) = k_0$ and $\kappa_j(T) = 0$ for the terminal time T, despite a_j having admissible pending requests throughout $[t_2, T]$.

Agent a_i has consumed all available capacity in the relevant context, leaving a_j unable to execute.

Temporal domination captures the first-mover advantage that arises when a shared resource is subject to depletion. It is distinct from Sybil amplification: it can occur with a single actor per agent and without any identity fragmentation.

4.3 Resource Contention Unfairness

Definition 4.4 (Resource Contention Unfairness)**.** For resource $r \in \mathcal{R}$, let $S_i^{(r)}$ denote the fraction of admitted actions on r attributable to agent a_i. A system exhibits *resource contention unfairness* at resource r if, in the absence of an explicit allocation policy for r, the distribution $\{S_i^{(r)}\}$ exhibits persistent skew:

$$\exists\, a_i, a_j \in \mathcal{A},\ r \in \mathcal{R} : \quad \left|S_i^{(r)} - S_j^{(r)}\right| > \varepsilon$$

due solely to non-identity factors such as request timing, retry rate, or network latency—not to differences in admissibility.

Table 1: Failure modes and the fairness properties they violate. A checkmark indicates the mode is consistent with the property; a cross indicates violation.

Failure Mode	**Share Fair**	**Actor Prop.**	**Strategy-Proof**
Sybil amplification	✓	×	×
Temporal domination	×	×	✓
Resource contention	×	×	✓

5 Main Results

5.1 Sybil Amplification Theorem

Theorem 5.1 (Sybil Amplification)**.** *Let $\mathcal{A}$ be a population of N agents with per-agent bound k_0, global capacity $K = N \cdot k_0$, and actor mapping $\pi : \mathcal{A} \to \mathcal{U}$. Let the allocation function α be identity-oblivious (i.e., α does not use π). Then for any actor u_j controlling m_j agents:*

(i) The actor share satisfies $S_{u_j} = m_j/N$ under exact agent-level share fairness.

(ii) There exists an assignment of m_j and N such that S_{u_j} is arbitrarily close to 1.

(iii) No identity-oblivious allocation mechanism can guarantee ε-actor-level proportionality for $\varepsilon < \max_j(m_j/N - 1/M)$.

Proof. **(i).** Under exact agent-level share fairness (Definition 3.1 with $\varepsilon = 0$), each agent a_i executes exactly k_0 admitted actions. Therefore:

$$S_{u_j} \;=\; \sum_{a_i \in \mathcal{A}_j} S_i \;=\; \sum_{a_i \in \mathcal{A}_j} \frac{k_0}{N \cdot k_0} \;=\; \frac{m_j \cdot k_0}{N \cdot k_0} \;=\; \frac{m_j}{N}.$$

(ii). Set $m_j = N - 1$ (actor u_j controls all but one agent) and N arbitrarily large. Then:

$$S_{u_j} \;=\; \frac{N-1}{N} \;\to\; 1 \quad \text{as } N \to \infty.$$

(iii). The deviation from actor-level proportionality for actor u_j under exact share fairness is:

$$\left|S_{u_j} - \tfrac{1}{M}\right| \;=\; \left|\tfrac{m_j}{N} - \tfrac{1}{M}\right|.$$

When the agent population is skewed (actors control unequal numbers of agents), $\max_j |m_j/N - 1/M| > 0$. Since α is identity-oblivious, it cannot distinguish agents by actor; therefore it cannot compensate for this skew. An identity-oblivious mechanism that achieves ε-actor-level proportionality must achieve $\varepsilon \geq \max_j |m_j/N - 1/M|$. For any fixed $\varepsilon < \max_j(m_j/N - 1/M)$, no identity-oblivious α satisfies the bound. □

Corollary 5.2 (Linear scaling of actor capacity)**.** *Under the conditions of Theorem 5.1, each additional agent registered by actor u_j increases S_{u_j} by exactly $1/N$. The actor share scales linearly with the number of controlled agents: $\partial S_{u_j}/\partial m_j = 1/N > 0$.*

Corollary 5.2 formalizes the observation from Example 1.1: ACP's Experiment 13 coordination window $\mathrm{CW}_{\mathrm{appr}} = 2N$ is not a flaw in ACP's design—it is the correct consequence of independent per-agent enforcement. The gap is at the allocation layer, not the admission layer.

5.2 Allocation Necessity Theorem

Theorem 5.3 (Allocation Necessity)**.** *There exists a multi-agent system* $(\mathcal{A}, F, \alpha_\emptyset)$ *in which* F *satisfies the atomic decision boundary property of Fernandez [5] and* $\alpha_\emptyset$ *is the trivial first-come-first-served allocation, such that:*

(i) Every admission decision by F *is atomically correct:* $\mathrm{Adm}(s, x) = \mathsf{true}$ *whenever* $F(s, x) = (\textsc{Allow}, s')$.

(ii) The system violates actor-level proportionality with arbitrarily large deviation $|S_{u_j} - 1/M| \to 1 - 1/M$.

Consequently, atomic correctness of F *does not imply any fairness property. An explicit allocation mechanism satisfying Definition 3.2 is a* necessary additional component *of a fair governance system.*

Proof. **Constructing the system.** Let $\mathcal{A} = \{a_1, a_2\}$, $\mathcal{U} = \{u_1, u_2\}$, $\pi(a_1) = u_1$, $\pi(a_2) = u_2$, and $k_0 = T$ for large T. Define:

$$\begin{aligned} S &= \{s_0, s_1, \ldots\}, \\ \mathrm{Adm}(s, x) &= \mathsf{true} \quad \forall s, x \text{ (all actions admissible everywhere)}, \\ F(s, x) &= (\textsc{Allow}, T(s, x)) \quad \forall s, x. \end{aligned}$$

The atomic boundary F is vacuously correct: it approves every action in every state, consistently with Adm.

Allocation. Let $\alpha_\emptyset$ select requests in the order they arrive. Suppose a_1 submits T requests at time $t = 0$ and a_2 submits T requests at time $t = 1$. Under FCFS, a_1's T requests are all processed before a_2's requests are reached. With $k_0 = T$, the counter $\kappa_1 = T$ is reached after a_1's requests; a_2 receives $\kappa_2 = 0$.

Fairness failure. $S_{u_1} = T/T = 1$ and $S_{u_2} = 0/T = 0$. Actor-level proportionality requires $|S_{u_j} - 1/2| \le \varepsilon$ for $M = 2$; here $|S_{u_1} - 1/2| = 1/2$, achievable as a worst case for any $\varepsilon < 1/2$.

Atomic correctness. Every action approved by F satisfies $\mathrm{Adm}(s, x) = \mathsf{true}$—the atomic boundary guarantee of Fernandez [5], Definition 3.3, is satisfied.

Necessity. Since the violation occurs for the trivial FCFS $\alpha_\emptyset$, and F alone has no mechanism to enforce distributional constraints over α's decisions, a fair outcome requires an α that explicitly enforces Definition 3.2. No property of F can substitute for this. □

5.3 Strategy-Proofness Impossibility

Theorem 5.4 (Strategy-Proofness Impossibility under Per-Agent Enforcement)**.** *Let* α *be an identity-oblivious allocation mechanism. Then* α *cannot simultaneously satisfy:*

(i) ε*-actor-level proportionality for any* $\varepsilon < \max_j |m_j/N - 1/M|$*; and*

(ii) strategy-proofness against identity fragmentation (Definition 3.5).

Any actor-aware mechanism that achieves both properties must use π *to aggregate state across agents of the same actor, and thereby breaks per-agent independence.*

Proof. **(Identity-oblivious mechanisms fail (i)).** By Theorem 5.1(iii), any identity-oblivious α cannot guarantee ε-actor-level proportionality when $m_j \neq N/M$. Since actor u_j can set m_j

arbitrarily (the mechanism cannot prevent agent registration), condition (i) is violated for any fixed ε.

(Actor-aware mechanisms fail (ii)). Suppose α is actor-aware and satisfies actor-level proportionality at $\varepsilon = 0$: each actor receives share $1/M$. Consider actor u_j splitting its single agent a_j into two agents $a_j^{(1)}, a_j^{(2)}$. Under actor-level proportionality, S_{u_j} remains $1/M$ regardless of m_j—the split does not help. Strategy-proofness (Definition 3.5) requires $\partial S_{u_j}/\partial m_j = 0$, which is satisfied. However, achieving this requires the mechanism to track $\mathcal{A}_j = \pi^{-1}(u_j)$ across all agents and aggregate their counters into a shared per-actor quota. This violates per-agent independence: the counter κ_i for agent a_i must now reference the aggregate state of $\mathcal{A}_j$, not just a_i's local history.

(Joint impossibility under per-agent independence). If α must be identity-oblivious (per-agent independent), then by the first part it cannot satisfy (i). If it sacrifices identity-oblivious, it must aggregate actor state—breaking per-agent independence. Therefore, no mechanism can simultaneously achieve per-agent independence, actor-level proportionality, and strategy-proofness. □

Remark 5.5 (Practical consequence). Theorem 5.4 establishes a fundamental three-way tradeoff: *per-agent independence*, *actor-level proportionality*, and *strategy-proofness against fragmentation* are mutually incompatible. Any governance system must sacrifice at least one. ACP [6] in its current form (ACP-RISK-3.0) sacrifices actor-level proportionality: it achieves per-agent independence via *PatternKey*-scoped counters and is strategy-proof within each scope, but does not bound actor-level share. The allocation mechanisms in §6 navigate this tradeoff differently.

6 Allocation Mechanisms

6.1 M1: Per-Agent Token Bucket (ACP Baseline)

The per-agent token bucket is the allocation mechanism implicit in ACP [6]. Each agent a_i maintains an independent counter κ_i bounded by k_0. Requests are processed in arrival order; there is no cross-agent coordination.

Definition 6.1 (Per-Agent Token Bucket)**.** The allocation function α_{TB} selects the earliest-arriving pending request among all agents with $\kappa_i < k_0$. If no agent has remaining quota, requests are queued.

M1 is correct by construction: it relies on F's atomic boundary for admission decisions and adds no allocation layer above it. By Theorem 5.1, it fails actor-level proportionality and is not strategy-proof against fragmentation.

6.2 M2: Round-Robin Fair Queuing

Definition 6.2 (Round-Robin Fair Queuing)**.** The allocation function α_{RR} serves pending requests in cyclic order over agents: at each step, it selects the next agent in a fixed rotation $\sigma = (a_{\sigma(1)}, a_{\sigma(2)}, \ldots, a_{\sigma(N)})$, skipping agents with no pending request or exhausted quota.

M2 achieves exact 0-share fairness and starvation freedom (every agent with a pending request eventually reaches the front of the rotation). It is identity-oblivious and therefore, by Theorem 5.1, fails actor-level proportionality.

6.3 M3: Actor-Aware Rate Limiting

Definition 6.3 (Actor-Aware Rate Limiting)**.** Let $K_U = K/M$ be the per-actor quota. The allocation function α_{AA} maintains a per-actor counter $\kappa_{u_j} = \sum_{a_i \in \mathcal{A}_j} \kappa_i$ and refuses to process requests from agents of u_j once $\kappa_{u_j} \geq K_U$. Among actors below quota, requests are served in round-robin order.

M3 achieves actor-level proportionality and strategy-proofness: an actor gains nothing by registering additional agents because the per-actor counter κ_{u_j} is shared across all agents in $\mathcal{A}_j$. The cost is that M3 requires knowledge of π and breaks per-agent independence.

6.4 M4: Weighted Fair Queuing

Definition 6.4 (Weighted Fair Queuing)**.** Assign a weight $w_i > 0$ to each agent a_i. The allocation function α_{WFQ} serves requests in proportion to weights: over any sufficiently long interval, the fraction of ALLOW decisions received by a_i is $w_i / \sum_j w_j$. (The precise bit-by-bit WFQ algorithm is due to Parekh and Gallager [11]; we use the proportional guarantee as the defining property.)

WFQ generalizes M2 (uniform weights) and M3 (set $w_i = 1/m_j$ for all $a_i \in \mathcal{A}_j$ to equalize actor shares). When weights are set to $1/m_j$ for each agent, α_{WFQ} achieves actor-level proportionality without requiring an explicit per-actor counter, but requires knowledge of m_j (and hence π) to compute the weights.

Table 2: Fairness properties achieved by each allocation mechanism. ✓= guaranteed; ≈ = achieved in expectation or asymptotically; × = not guaranteed.

Mechanism	**Share Fair**	**Actor Prop.**	**Strategy-Proof**	**Starvation-Free**
M1 Token Bucket	≈	×	×	✓
M2 Round-Robin	✓	×	×	✓
M3 Actor-Aware	✓	✓	✓	✓
M4 WFQ (uniform)	✓	×	×	✓
M4 WFQ (actor-wt)	≈	✓	✓	✓

Remark 6.5 (Impossibility of × entries for identity-oblivious mechanisms)*.* The × entries for M1, M2, and M4-uniform in the Actor Proportionality and Strategy-Proofness columns are not design deficiencies but structural impossibilities. Theorem 5.1(iii) proves that no identity-oblivious α can guarantee ε-actor-level proportionality for $\varepsilon < \max_j(m_j/N - 1/M)$. Since M1, M2, and M4-uniform do not use π, these entries are theorem consequences, not engineering gaps. Theorem 5.4 further proves that any mechanism simultaneously achieving actor-level proportionality and strategy-proofness must use π—which M1, M2, and M4-uniform do not—making the × entry for Strategy-Proofness of those mechanisms similarly unavoidable.

7 Analysis of Mechanisms

7.1 Correctness of M2 (Round-Robin)

Proposition 7.1 (M2 achieves share fairness and starvation freedom)**.** *Under α_{RR}, over any execution trace of length T with all agents continuously submitting requests, the share deviation satisfies $|S_i - 1/N| \leq 1/T$ for all a_i. Furthermore, every agent with a pending admissible request eventually receives an* ALLOW *decision.*

Proof. In each round of the rotation, each agent is served at most once. After k complete rounds, $\kappa_i \in \{k, k+1\}$ for all i, giving $|\kappa_i/K' - 1/N| \leq 1/(N \cdot k) \leq 1/T$ for $T = N \cdot k$. Starvation freedom follows because every agent is visited in every round; an agent with a pending request and remaining quota is served within at most N steps. □

7.2 Correctness of M3 (Actor-Aware)

Proposition 7.2 (M3 achieves actor-level proportionality and strategy-proofness)**.** *Under α_{AA}, for all actors $u_j \in \mathcal{U}$, the actor share satisfies $S_{u_j} \leq K_U/K' \leq 1/M$ regardless of m_j. Registering additional agents does not increase S_{u_j}.*

Proof. By construction, α_{AA} processes at most $K_U = K/M$ requests from actor u_j's agents in total, regardless of how many agents u_j controls. The per-actor counter κ_{u_j} is capped at K_U. Therefore $S_{u_j} = K_U/K' \leq 1/M$. An additional agent a_i^{new} under u_j does not increase K_U or reduce the cap: κ_{u_j} still saturates at K_U. Hence $\partial S_{u_j}/\partial m_j = 0$. □

Proposition 7.3 (M3 fails under misrepresented actor mapping)**.** *Suppose actor u_j controls m_j agents but presents a falsified mapping $\tilde{\pi}$ attributing each of its agents to a distinct phantom actor: $\tilde{\pi}(a_i) = \tilde{u}_i$ for m_j distinct identifiers $\tilde{u}_i \notin \mathcal{U}$. Under M3 applied to $\tilde{\pi}$, the aggregate quota available to u_j is $m_j \cdot K_U$, restoring full Sybil amplification:*

$$S_{u_j}(M3, \tilde{\pi}) \;=\; \frac{m_j}{M} \;\geq\; \frac{m_j}{N} \;\geq\; \frac{1}{M} \quad \textit{for } m_j > 1,$$

where the second inequality holds since $M \leq N$. In particular, $\partial\, S_{u_j}/\partial m_j = 1/M > 0$, violating strategy-proofness.

Proof. Under $\tilde{\pi}$, M3 treats each agent $a_i \in \mathcal{A}_j$ as belonging to a distinct actor $\tilde{u}_i$. Each phantom actor receives the full per-actor quota $K_U = K/M$. The aggregate quota available to u_j's m_j agents under $\tilde{\pi}$ is therefore $m_j \cdot K_U = m_j\, K/M$, giving actor share $S_{u_j} = m_j/M$. Since $M \leq N$, we have $m_j/M \geq m_j/N$, meaning misrepresentation under M3 is *strictly worse* than the unmitigated M1 case for any $M < N$: the actor captures more capacity per agent than it would under an identity-oblivious mechanism. The derivative $\partial(m_j/M)/\partial m_j = 1/M > 0$ violates strategy-proofness (Definition 3.5) even more severely than Theorem 5.1(i) predicts for M1. □

Remark 7.4 (Observability requirement for actor-aware mechanisms). Proposition 7.3 formalizes the assumption implicit in M3 and M4 (actor-wt): the actor mapping π must be verifiable by the allocation mechanism. Without a trusted identity layer—such as a cryptographically signed registration authority or a hardware-attestation scheme—actor-aware mechanisms are no stronger than identity-oblivious mechanisms in the worst case, and can be strictly worse under active misrepresentation (since $m_j/M \geq m_j/N$). The design choice between M1/M2 and M3/M4 therefore carries an implicit prerequisite: identity observability must be guaranteed at the infrastructure level before actor-aware allocation can be deployed.

7.3 Coordination Cost

Proposition 7.5 (Actor-aware mechanisms require shared actor state)**.** *Any allocation mechanism satisfying actor-level proportionality must maintain, for at least one actor u_j with $m_j > 1$, a shared state variable that aggregates the admission counters of agents in $\mathcal{A}_j$. This requires $O(M)$ shared counters updated at every* Allow *decision.*

Proof. Suppose the mechanism achieves actor-level proportionality but no agent in $\mathcal{A}_j$ can observe the counters of its sibling agents. Then agents in $\mathcal{A}_j$ operate independently: each sees only its own κ_i. They cannot collectively halt when $\kappa_{u_j} = K_U$ unless some shared signal is propagated. To propagate this signal, the system must maintain a counter κ_{u_j} shared across $\mathcal{A}_j$ and increment it atomically with each ALLOW issued to any $a_i \in \mathcal{A}_j$. This requires one shared counter per actor, $O(M)$ total. □

Proposition 7.5 makes precise the practical cost of actor-level fairness: it requires $O(M)$ additional shared state in the admission control infrastructure, and each ALLOW decision must atomically update both the per-agent counter κ_i and the per-actor counter κ_{u_j}. This is achievable within the evaluate-then-mutate execution contract of ACP without violating the atomic boundary: the atomic step F can jointly update κ_i, κ_{u_j}, and the ledger state in one transaction.

7.4 Asymptotic Guarantee of M4 (WFQ with Actor Weights)

Proposition 7.6 (M4 achieves actor-level proportionality asymptotically)**.** *Under α_{WFQ} with weights $w_i = 1/m_j$ for all $a_i \in \mathcal{A}_j$, the actor share satisfies:*

$$\left| S_{u_j}(T) - \tfrac{1}{M} \right| \;\leq\; \frac{C_{\mathrm{WFQ}}}{T}$$

for a constant C_{WFQ} depending only on N, M, and the maximum weight ratio $w_{\max}/w_{\min}$, for all T sufficiently large.

Proof. Under WFQ with weights $\{w_i\}$, the fractional allocation to agent a_i over any window $[t, t{+}T]$ satisfies $|\kappa_i(T)/T - w_i/W| \leq C^{(i)}_{\mathrm{WFQ}}/T$ where $W = \sum_j w_j$ (this is the standard WFQ bitlag bound of Parekh and Gallager [11] applied to the admission counter). Summing over $a_i \in \mathcal{A}_j$:

$$S_{u_j}(T) \;=\; \sum_{a_i \in \mathcal{A}_j} \frac{\kappa_i(T)}{K'(T)} \;\to\; \sum_{a_i \in \mathcal{A}_j} \frac{w_i}{W} \;=\; \frac{m_j \cdot (1/m_j)}{W} \;=\; \frac{1}{W}$$

as $T \to \infty$. With M actors each receiving weight sum 1, $W = M$, giving $S_{u_j}(\infty) = 1/M$. The finite-T deviation is bounded by C_{WFQ}/T where $C_{\mathrm{WFQ}} = \sum_{a_i \in \mathcal{A}_j} C^{(i)}_{\mathrm{WFQ}}$. □

Remark 7.7 (M4 vs. M3: the identity-observability tradeoff)*.* Proposition 7.6 shows that M4 achieves actor-level proportionality *asymptotically* ($O(1/T)$ convergence), whereas M3 achieves it *exactly* in the worst case. The practical tradeoff is one of state architecture: M3 requires a per-actor shared counter κ_{u_j} that must be atomically incremented; M4 requires per-agent weights that must be set at registration time (and recomputed when actors add or remove agents). In systems where actor population is stable, M3 is simpler and stronger. In systems with frequent agent churn, M4's weight recomputation may be more tractable than maintaining exact per-actor counters across dynamic membership.

8 Interaction with Escalation and IML

8.1 Escalation Queue Fairness

The ESCALATE outcome of Fernandez [5] suspends a state transition pending supervisor resolution. In a multi-agent system, multiple agents may escalate simultaneously, creating a *pending-review queue* $P \subseteq S \times A_{\mathrm{ag}}$ (Definition 3.6 of Paper 0). The Escalation Closure Requirement (Corollary 5.6

of Paper 0) demands that each resolution be atomic. We add a complementary requirement: the order in which pending escalations are resolved must itself be fair.

Definition 8.1 (Escalation Queue Fairness)**.** The pending-review queue P is governed by an *escalation allocation function* $\alpha_P : \mathcal{P}(P) \times S \to P$ that selects the next pending request for resolution. The system satisfies *escalation queue fairness* if α_P satisfies the same share fairness and actor-level proportionality properties (Definitions 3.1–3.2) with respect to the escalation queue.

An actor that escalates requests faster than others can, under FCFS resolution, dominate supervisor attention—analogous to temporal domination at the admission boundary. Actor-aware escalation queuing (M3 applied to α_P) prevents this.

8.2 Allocation Bias as Behavioral Drift

A persistent skew in the allocation of admissible actions is a *distributional* property of the agent population. We connect it to the IML framework of Fernandez [7].

Definition 8.2 (Allocation Bias Signal)**.** For agent a_i with admission-time baseline share $S_i^{(0)} = 1/N$ (assuming equal initial conditions), the *allocation bias* at time t is:

$$B_i(t) \;=\; S_i(t) - \tfrac{1}{N}.$$

The *portfolio deviation* for actor u_j is:

$$\Delta_{u_j}(t) \;=\; \sum_{a_i \in \mathcal{A}_j} \big|\, B_i(t) \,\big|,$$

measuring the aggregate allocation skew attributable to u_j's agents.

Theorem 8.3 (Allocation bias induces IML drift)**.** *Let $a_i \in \mathcal{A}$ be an agent with admission-time snapshot $\mathcal{A}_0^{\text{emp}} = (P_{E_0}, \mu_{\text{depth}}, \sigma_{\text{depth}})$, calibrated during burn-in under fair allocation ($B_i = 0$). Let P_{excess} denote the distribution of tool calls drawn from states visited only because of excess allocation, and suppose $\text{JS}(P_{\text{excess}} \| P_{E_0}) \geq \eta > 0$. If the agent is persistently over-served with $B_i(t) \geq \delta > 0$ over a window of T steps, let $f = \delta \,/\, (\frac{1}{N} + \delta)$ be the fraction of actions attributable to excess allocation. Then:*

$$\widehat{D}(\tau_i,\, \mathcal{A}_0) \;\geq\; 0.40 \cdot \frac{f^2\,\eta}{2} \;>\; 0.$$

Throughout, the enforcement signal satisfies $g(\tau_i) = 0$, since no individual action violates a local constraint.

Proof. By linearity of empirical distributions, after T steps the empirical tool distribution of a_i decomposes as:

$$\hat{P}_{\tau_i} \;=\; (1-f)\,\hat{P}_{\text{base}} \;+\; f\,P_{\text{excess}},$$

where $\hat{P}_{\text{base}}$ converges to P_{E_0} as $T \to \infty$ under the bounded-drift Assumption A1 of Fernandez [7], since the base fraction of actions is drawn from the same distribution as during burn-in. By joint convexity of Jensen–Shannon divergence and the inequality

$$\text{JS}(\lambda P + (1-\lambda) Q \parallel P) \;\geq\; \tfrac{\lambda^2}{2}\,\text{JS}(Q \| P), \quad \lambda \in [0,1],$$

applied with $\lambda = f$, $P = \hat{P}_{\text{base}} \approx P_{E_0}$, and $Q = P_{\text{excess}}$:

$$\text{JS}(\hat{P}_{\tau_i} \| P_{E_0}) \;\geq\; \frac{f^2}{2}\,\text{JS}(P_{\text{excess}} \| P_{E_0}) \;\geq\; \frac{f^2\,\eta}{2}.$$

Since the temporal drift component of Fernandez [7], Definition 4.1, is $D_t(\tau_i) = \mathrm{JS}(\hat{P}_{\tau_i} \| P_{E_0})$, and the IML deviation satisfies $\widehat{D} = 0.40\, D_t + 0.35\, D_c + 0.25\, D_l \geq 0.40\, D_t$, we obtain $\widehat{D} \geq 0.40 \cdot f^2 \eta / 2 > 0$.

The enforcement signal $g(\tau_i) = 0$ throughout because P_{excess} consists of individually admissible actions (each passes the point-wise rule evaluation of Assumption 2.1 of Fernandez [7]). The IML therefore detects drift while enforcement remains silent—an instance of the Non-Identifiability regime (Theorem 3.2 of Fernandez [7]) triggered structurally by the allocation layer rather than by a change in the agent's intrinsic behavior. □

Corollary 8.4 (Detection delay amplification under under-service)**.** *Let a_i receive persistent under-service $B_i(t) \leq -\delta < 0$, so its effective execution rate is $r = \frac{1}{N} - \delta < \frac{1}{N}$. The IML detection delay for genuine behavioral drift in a_i, as given by Theorem 3.4 of Fernandez [7], satisfies:*

$$T^*_{\text{under}}(\theta) \;\geq\; \frac{T^*_0(\theta)}{1 - \delta N},$$

where $T^*_0(\theta)$ is the detection delay under fair allocation. As $\delta \nearrow 1/N$ (complete starvation), $T^*_{\text{under}} \to \infty$.

Proof. The detection delay bound $T^*_0(\theta) = t_0 + \lceil c_0/(p_1^2 \ln(1/\alpha)) \rceil$ of Fernandez [7] is derived with agent a_i executing at rate $\frac{1}{N}$ of total capacity, yielding sample count $n_0 = T/N$ at time T. Under under-service at rate $r = \frac{1}{N} - \delta$, the sample count after T steps is $n_r = r\,T = (1 - \delta N)\,T/N$. Since estimation error $\varepsilon_{\text{est}}(n)$ is non-increasing in n (Assumption A4 of Fernandez [7]), fewer samples inflate the effective detection delay by $n_0/n_r = 1/(1 - \delta N)$, giving $T^*_{\text{under}} \geq T^*_0/(1 - \delta N)$. As $\delta \to 1/N$, the denominator $1 - \delta N \to 0$, so $T^*_{\text{under}} \to \infty$: a fully starved agent is unmonitorable by IML. □

Theorem 8.3 establishes a precise bridge between the allocation layer and the IML: Sybil amplification that produces persistent allocation bias is detectable by the IML as a portfolio-level drift signal, even though no individual admission decision is incorrect. Corollary 8.4 shows the dual failure: under-served agents are harder to monitor, as their reduced sample count inflates detection delay proportionally. A well-calibrated allocation mechanism therefore prevents two distinct IML failure modes simultaneously—false positives from over-service and detection blindness from under-service.

8.3 BAR-Monitor Interference under Throttled Allocation

The Boundary Activation Rate (BAR-Monitor) of Fernandez [6] detects deviation collapse by measuring the fraction of admission requests that trigger Escalate or Refuse decisions at F. Its invariant presupposes that all requests *reach the boundary*. Under actor-aware allocation (M3), requests from quota-exhausted actors are intercepted at the allocation layer and never presented to F—creating a systematic downward bias in BAR that can mimic deviation collapse while enforcement remains fully functional.

Proposition 8.5 (M3 reduces BAR below its correct baseline)**.** *Let $u_j \in \mathcal{U}$ have reached its per-actor quota $K_U = K/M$, so that α_{AA} drops all subsequent requests from agents in $\mathcal{A}_j$ before they reach F. Let $\mathrm{BAR}_N^{(0)}$ denote the Boundary Activation Rate under M2 (all agents served), and $\mathrm{BAR}_N^{(\text{M3})}$ the rate under M3. Then:*

$$\mathrm{BAR}_N^{(\text{M3})} \;\leq\; \mathrm{BAR}_N^{(0)} \cdot \Big(1 - \tfrac{m_j}{N}\Big).$$

If $m_j/N > 1 - \theta_{\text{collapse}}/\text{BAR}_N^{(0)}$, the BAR-Monitor of Fernandez [6] enters a false low-activation regime even though F is not in deviation collapse.

Proof. Under M2, all N agents present requests to F in round-robin order. The activation count in window N is $A = |\{d_i : d_i \in \{\textsc{Escalate}, \textsc{Refuse}\}\}|$, giving $\text{BAR}_N^{(0)} = A/N$. Under M3 with u_j at quota, the m_j agents of u_j are blocked at the allocation layer; only $N - m_j$ agents reach F. The window of N evaluation slots now contains at most $N - m_j$ evaluations by F (the m_j throttled slots are never forwarded). Since activation events occur only among requests that reach F, and F's behavior on the remaining $N - m_j$ agents is unchanged:

$$\text{BAR}_N^{(\text{M3})} = \frac{A'}{N} \leq \frac{A}{N} \cdot \frac{N - m_j}{N} = \text{BAR}_N^{(0)} \cdot \left(1 - \tfrac{m_j}{N}\right).$$

If this falls below θ_{collapse}, the BAR-Monitor issues a deviation collapse alert (low-activation regime, Definition 4.6 of Fernandez [6]). Yet F retains full enforcement capacity: were the throttled requests forwarded, F would correctly ESCALATE or REFUSE them. The alert is a false positive caused by allocation-layer throttling, not enforcement failure. □

Remark 8.6 (Corrected activation rate). A BAR-Monitor co-deployed with an actor-aware allocation layer must normalize by requests *presented to* F, not by total requests submitted. The corrected rate

$$\text{BAR}_N^{\text{corr}} = \frac{|\{d_i : d_i \in \{\textsc{Escalate}, \textsc{Refuse}\}\}|}{|\{r_i : r_i \text{ reached } F\}|}$$

is independent of which allocation mechanism throttles upstream, recovering the invariant $\text{BAR}_N^{\text{corr}} = \text{BAR}_N^{(0)}$ regardless of m_j or quota state.

8.4 The Four-Layer Governance Architecture

The four papers in this series define four distinct, non-redundant layers of governance. Figure 1 depicts their relationship. Each layer addresses a failure mode invisible to the layer below it: enforcement-time state changes (Paper 0), per-action admissibility (Paper 1), within-compliance behavioral drift (Paper 2), and cross-agent allocation skew (Paper 3).

The non-redundancy claim deserves emphasis. A system at Layer 1 satisfies the atomic boundary but may fail Layers 2–4. A system at Layer 2 (ACP) is atomically correct and enforces bounded per-action admission, but (by Theorem 5.3) may fail Layers 3–4. A system at Layer 3 (IML-equipped) monitors behavioral drift but does not address which agent gets access to the boundary (Layer 4). Only a system operating at all four layers simultaneously provides the full governance guarantee: no inadmissible action fires (Layers 1–2), behavioral drift from admission-time contracts is detected (Layer 3), and admissible actions are distributed fairly across agents and actors (Layer 4).

Crucially, each layer *assumes* the layer below it is functioning: IML takes atomicity as given; the allocation layer takes ACP enforcement as given. This layered dependency is not circular but *compositional*: each paper in the series contributes a component that is necessary but not sufficient alone, and the quartet is closed exactly when all four components are present.

9 Discussion and Related Work

9.1 Relationship to Classical Fairness Theory

Fair resource allocation has a long history in both economics and networking. *Max-min fairness* [1, 3] maximizes the minimum share across all agents, naturally preventing starvation. *Proportional*

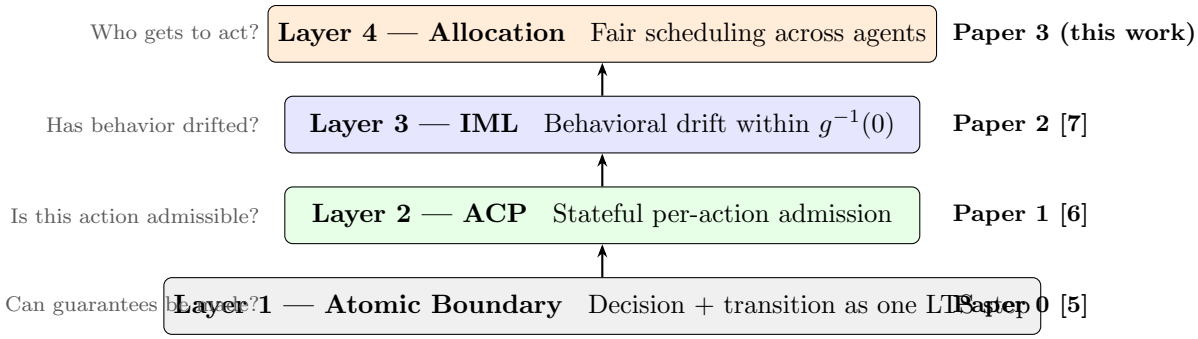

Figure 1: The four-layer governance architecture. Each layer addresses a governance failure invisible to the layer below. Layer 1 (Paper 0): structural impossibility of split systems; establishes the atomic boundary as the minimal correctness requirement. Layer 2 (Paper 1, ACP): per-action admission control enforced atomically via Execution Tokens. Layer 3 (Paper 2, IML): detection of behavioral drift within the compliance region $g^{-1}(0)$ where enforcement is structurally blind. Layer 4 (this paper): fair scheduling of which agent presents its request to the atomic boundary, preventing Sybil amplification and allocation-level concentration. The layers are non-redundant: correctness at Layer k does not imply correctness at Layer $k+1$.

fairness [10] maximizes the sum of logarithmic utilities, distributing capacity proportionally to demand. *Envy-freeness* [8, 13] requires that no agent prefers another's allocation.

Our work departs from classical fair-division theory in three respects. First, the resource being allocated is not a divisible good but *access to an indivisible atomic boundary*: each "quantum" of capacity is one evaluation step of F, and the boundary can only process one request at a time. Second, the correctness constraint (each admitted action must be admissible) interacts with the fairness constraint: we cannot simply round-robin all agents regardless of their admissibility state. Third, the actor structure (Definition 2.2) introduces an identity layer absent from classical models, giving rise to the Sybil amplification problem.

The strategy-proofness impossibility (Theorem 5.4) is a governance-layer analogue of the Gibbard–Satterthwaite theorem [9, 12]: just as no voting rule can be simultaneously non-dictatorial, Pareto-efficient, and strategy-proof, no per-agent-independent allocation mechanism can simultaneously guarantee actor-level proportionality and strategy-proofness against identity fragmentation.

9.2 Relationship to the Sybil Attack Literature

The Sybil attack was formally identified by Douceur [4] in the context of peer-to-peer reputation systems: a malicious node creates multiple fake identities to subvert voting or reputation aggregation. Identity-based defenses (trusted authorities, admission fees, resource costs) are the standard mitigations.

Our contribution is distinct: we provide a formal characterization of Sybil amplification within the LTS-based governance model, prove that it is consistent with full atomic correctness, and show that the mitigation (M3, actor-aware allocation) requires actor-state aggregation that trades off per-agent independence. Crucially, in our setting the "Sybil nodes" are not fake: they are legitimate agents operating within their individual bounds. The attack surface is not dishonesty but the structural separation between per-agent enforcement and system-wide allocation.

9.3 Relationship to Multi-Agent Resource Allocation

The formal study of multi-agent resource allocation [2] concerns the distribution of indivisible goods among self-interested agents, asking which allocations are efficient, envy-free, or Pareto-optimal. Our setting shares several features—multiple agents, limited capacity, competing interests—but departs in three important ways.

First, the *resource* in our model is not a good but an *access right to a governance mechanism*. The atomic boundary F is the resource being allocated; execution capacity is its product. Unlike goods, access rights have a correctness dimension: the allocation mechanism must respect the admissibility predicate Adm, which can refuse a request regardless of allocation priority. Fair-division theory does not include this constraint.

Second, our model is *sequential*: the allocation unfolds over time as agents submit requests and the LTS evolves. Fair-division results typically concern static one-shot assignments. The temporal failure modes we identify—temporal domination, starvation—are inherently dynamic and have no counterpart in static allocation theory.

Third, the actor-agent structure introduces a *strategic layer* absent from standard models: actors can manipulate their effective share by adjusting the number of agents they control, which is not a strategic variable in fair-division models where agents are primitive. The Gibbard–Satterthwaite analogue we prove (Theorem 5.4) addresses this strategic dimension and has no standard fair-division counterpart.

These three departures—correctness constraints, temporal dynamics, and strategic identity control—define the specific contribution of the present work relative to fair-division theory.

9.4 Position within the Governance Quartet

The four papers in this series form a logically closed architecture; Table 3 summarizes each paper's question, result, and the failure mode it addresses.

Table 3: The governance quartet: question, result, and failure mode addressed.

Paper	**Question**	**Central result**	**Failure mode**
0 [5]	When can admissibility be guaranteed?	Split systems cannot guarantee it; atomicity is necessary and sufficient.	TOCTOU / state change between decision and execution
1 [6]	How to implement the atomic boundary?	ACP: Execution Token issued and consumed atomically with ledger mutation.	Per-action correctness under concurrent agents
2 [7]	Does correct enforcement imply behavioral alignment?	No: $\mathcal{A}_0 \notin \sigma(g)$; IML restores observability.	Behavioral drift invisible to enforcement
3 (this work)	Does correct admission imply fair allocation?	No: Theorems 5.1–5.4; allocation layer is necessary.	Sybil amplification and cross-agent distribution failure

The papers are *non-redundant* in the sense that the result of each is invisible to the prior layers: admissibility failures in split systems (Paper 0) cannot be detected by policy engines that are split by construction; behavioral drift (Paper 2) cannot be detected by enforcement signals; allocation skew (this paper) cannot be detected by the IML, which monitors individual agents in isolation.

Together, they cover the full governance space from execution-time correctness to system-wide distributional fairness.

9.5 Limitations and Future Work

Two important questions are left for future work. First, we assume a fixed agent population; dynamic arrival and departure of agents introduces challenges for actor-aware mechanisms (how to reallocate K_U when actors join or leave) that are outside the scope of the present LTS model. Second, we analyze worst-case fairness; a stochastic model of request arrivals would permit average-case analysis and tighter bounds on the detection-delay amplification of Corollary 8.4.

The two questions previously listed as open—the interaction between M3 and ACP's BAR-Monitor, and the formal treatment of misrepresented actor mappings—have been resolved in §8.3 (Proposition 8.5 and Remark 8.6) and in the M3 analysis (Proposition 7.3 and Remark 7.4), respectively.

10 Conclusion

We have introduced the *allocation layer* as a necessary addition to atomic governance architectures. Using the LTS framework of Fernandez [5], we proved that a system can be atomically correct at every admission decision while simultaneously failing at the distribution of those decisions across agents. Correctness is a local property of state transitions; fairness is a global property of scheduling patterns. Neither implies the other.

The Sybil Amplification Theorem establishes the precise mechanism of failure: per-agent bounded enforcement scales linearly with agent count, enabling an actor to multiply its governance capacity without bound by registering additional agents. The Allocation Necessity Theorem establishes that no property of the atomic boundary function F can prevent this—a separate allocation mechanism above the boundary is required. The Strategy-Proofness Impossibility establishes the fundamental tradeoff: per-agent independence, actor-level proportionality, and resistance to identity fragmentation cannot all be achieved simultaneously.

The four allocation mechanisms we analyze navigate this tradeoff from different positions. Simple mechanisms (M1, M2) preserve per-agent independence but are Sybil-vulnerable. Actor-aware mechanisms (M3, M4 with actor weights) achieve proportionality and strategy-proofness but require cross-agent state aggregation and observability of the actor mapping. Deployers of multi-agent governance infrastructure must choose based on whether π is observable and whether identity fragmentation is a realistic threat in their environment.

This paper closes the governance quartet. Paper 0 [5] establishes that admissibility can only be guaranteed by atomic enforcement: the decision boundary is a structural necessity, not an engineering choice. Paper 1 [6] instantiates the boundary as ACP and demonstrates its correctness properties across thousands of states. Paper 2 [7] shows that enforcement-correct behavior can still drift from admission-time contracts, and introduces IML to detect this drift. The present paper shows that a drift-monitored, atomically correct system can still allocate its governance capacity unfairly—and provides the formal tools to characterize, prevent, and detect this failure.

> *Admission control determines what can happen. Fairness determines who gets to act. Correctness is local. Fairness is global. Both are necessary for governed autonomy.*

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